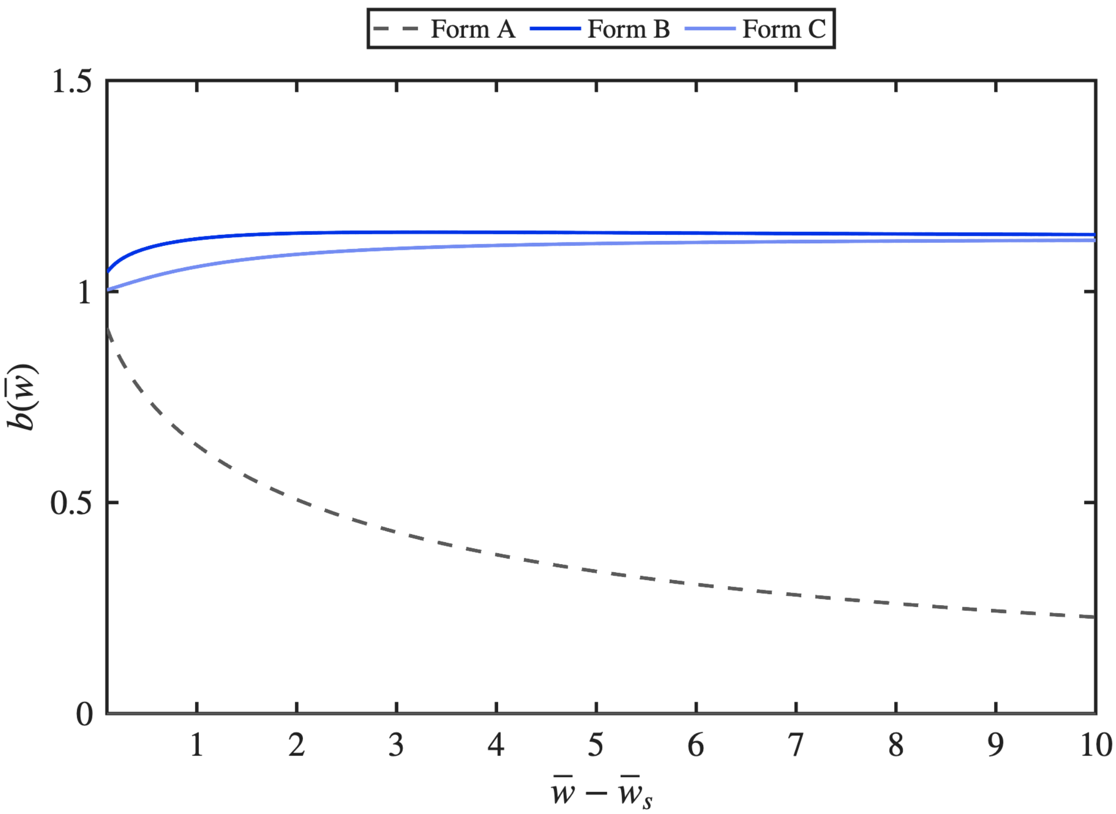
$$
\begin{array}{lll}
\text{Form A} & \bar{a} = 1 - e^{-b\left(\bar{w}-\bar{w}_s\right)} & p = 0 \\
\text{Form B} & \bar{a} = 1 - \dfrac{b}{\left(\bar{w}-\bar{w}_s\right) + b} & p = 1 \\
\text{Form C} & \bar{a} = 1 - \dfrac{b}{1+\bar{w}-\bar{w}_s} & p = 1
\end{array}
$$

$$
\bar{w}_s \text{ known}, \qquad \bar{a}'_{\text{true}} := \frac{\mathrm{d}\bar{a}_{\text{true}}}{\mathrm{d}\bar{w}}
$$

$$
b_{\mathrm{A}}(\bar{w}) = -\frac{\ln\left(1-\bar{a}_{\text{true}}\right)}{\bar{w}-\bar{w}_s}, \qquad b_{\mathrm{B}}(\bar{w}) = \left(\bar{w}-\bar{w}_s\right)\frac{1-\bar{a}_{\text{true}}}{\bar{a}_{\text{true}}}, \qquad b_{\mathrm{C}}(\bar{w}) = \left(1+\bar{w}-\bar{w}_s\right)\left(1-\bar{a}_{\text{true}}\right)
$$

$$
\bar{w}-\bar{w}_s \in [\,0.1,\ 10\,]
$$

$$
\begin{array}{rl}
\bar{w}-\bar{w}_s = 0: & \text{A, B}: \ \text{any } b\,; \qquad \text{C}: \ b = 1 \\
\bar{w}-\bar{w}_s = \infty: & \text{A, B, C}: \ \text{any } b \\
\text{elsewhere}: & \partial\bar{a}/\partial b \neq 0 \ \Rightarrow \ b(\bar{w}) \text{ unique}
\end{array}
$$

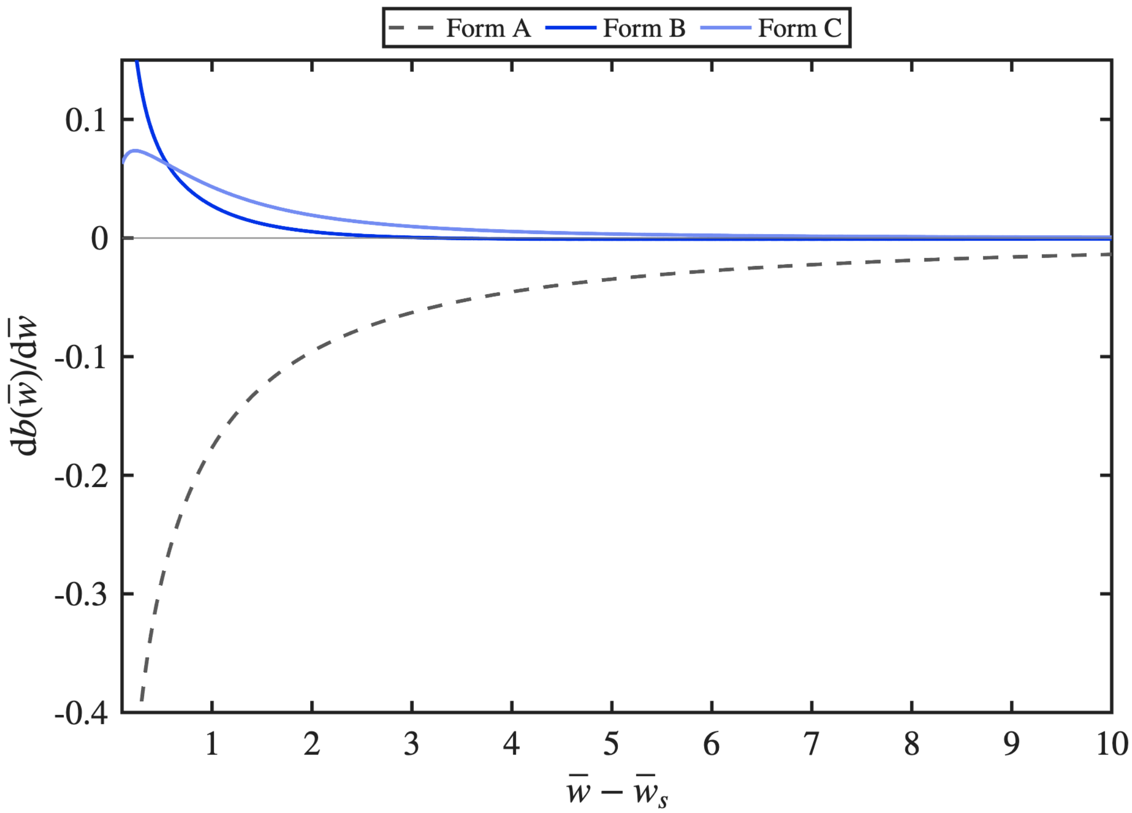


$$
\frac{\mathrm{d}b_{\mathrm{A}}(\bar{w})}{\mathrm{d}\bar{w}} = \frac{1}{\left(\bar{w}-\bar{w}_s\right)^2}\left[\frac{\left(\bar{w}-\bar{w}_s\right)\bar{a}'_{\mathrm{true}}}{1-\bar{a}_{\mathrm{true}}} + \ln\!\left(1-\bar{a}_{\mathrm{true}}\right)\right]
$$

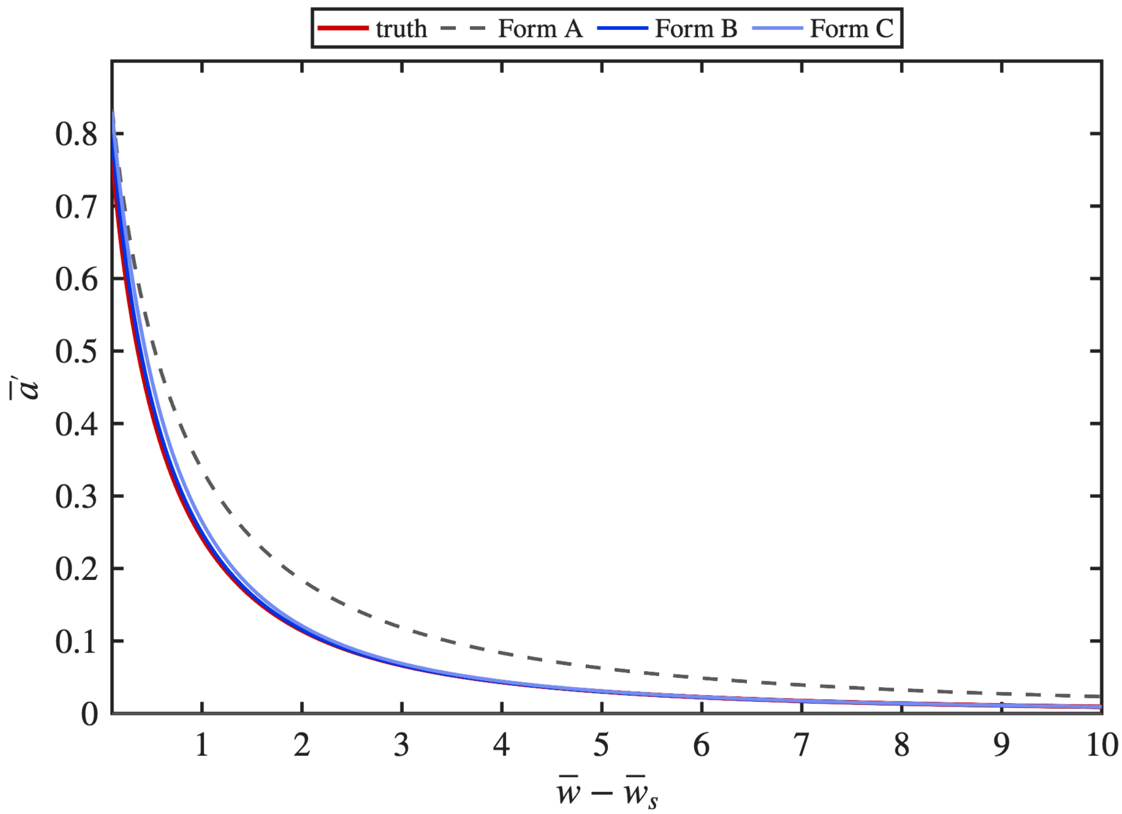
$$
\frac{\mathrm{d}b_{\mathrm{B}}(\bar{w})}{\mathrm{d}\bar{w}} = \frac{1-\bar{a}_{\mathrm{true}}}{\bar{a}_{\mathrm{true}}} - \left(\bar{w}-\bar{w}_s\right)\frac{\bar{a}'_{\mathrm{true}}}{\bar{a}^2_{\mathrm{true}}}
$$

$$
\frac{\mathrm{d}b_{\mathrm{C}}(\bar{w})}{\mathrm{d}\bar{w}} = \left(1-\bar{a}_{\mathrm{true}}\right) - \left(1+\bar{w}-\bar{w}_s\right)\bar{a}'_{\mathrm{true}}
$$

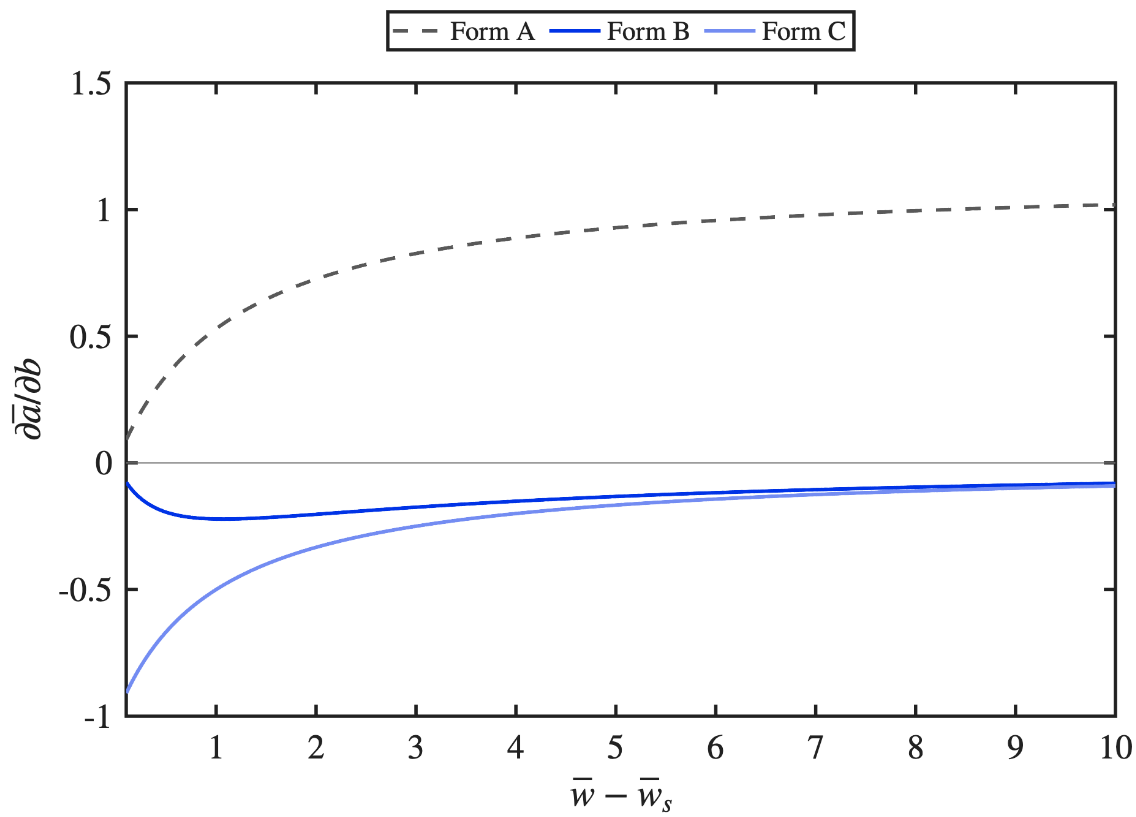
$$
\left.\frac{\mathrm{d}b(\bar{w})}{\mathrm{d}\bar{w}}\right|_{\bar{w}-\bar{w}_s=0.1} = -0.669,\ +0.316,\ +0.063 \qquad \left.\frac{\mathrm{d}b(\bar{w})}{\mathrm{d}\bar{w}}\right|_{\bar{w}-\bar{w}_s=10} = -0.014,\ -0.001,\ +0.001
$$



$$\bar{a}'_{\mathrm{A}} = -\frac{\left(1-\bar{a}_{\mathrm{true}}\right)\ln\left(1-\bar{a}_{\mathrm{true}}\right)}{\bar{w}-\bar{w}_s} \qquad \bar{a}'_{\mathrm{B}} = \frac{\bar{a}_{\mathrm{true}}\left(1-\bar{a}_{\mathrm{true}}\right)}{\bar{w}-\bar{w}_s} \qquad \bar{a}'_{\mathrm{C}} = \frac{1-\bar{a}_{\mathrm{true}}}{1+\bar{w}-\bar{w}_s}$$

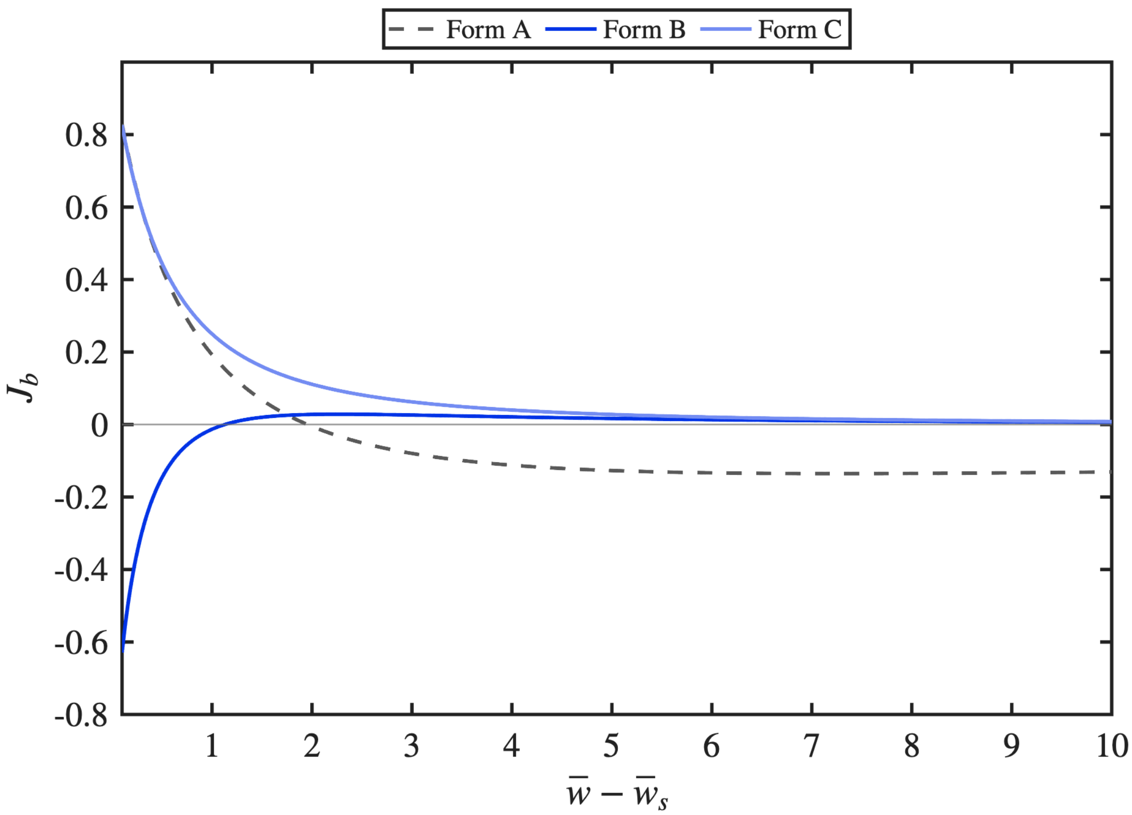


$$\frac{\partial \bar{a}_{\mathrm{A}}}{\partial b} = +\big(\bar{w} - \bar{w}_s\big)\big(1 - \bar{a}_{\mathrm{true}}\big) \qquad \frac{\partial \bar{a}_{\mathrm{B}}}{\partial b} = -\frac{\bar{a}_{\mathrm{true}}^2}{\bar{w} - \bar{w}_s} \qquad \frac{\partial \bar{a}_{\mathrm{C}}}{\partial b} = -\frac{1}{1 + \bar{w} - \bar{w}_s}$$



$$J_{b,\mathrm{A}} = \big(1-\bar{a}_{\mathrm{true}}\big)\Big[1+\ln\big(1-\bar{a}_{\mathrm{true}}\big)\Big] \qquad J_{b,\mathrm{B}} = \frac{\bar{a}_{\mathrm{true}}^2\big(2\bar{a}_{\mathrm{true}}-1\big)}{\big(\bar{w}-\bar{w}_s\big)^2} \qquad J_{b,\mathrm{C}} = \frac{1}{\big(1+\bar{w}-\bar{w}_s\big)^2}$$

$$J_b = 0: \quad \mathrm{A}:\ \bar{a}_{\mathrm{true}} = 1-e^{-1},\ \ \bar{w}-\bar{w}_s = 1.955 \qquad \mathrm{B}:\ \bar{a}_{\mathrm{true}} = \tfrac{1}{2},\ \ \bar{w}-\bar{w}_s = 1.128 \qquad \mathrm{C}:\ \text{none}$$



anchor $\quad b_0 = \frac{9}{8}\ (\perp), \quad \frac{9}{16}\ (\parallel)$ $\qquad$ far-field reflection coefficient, one number, fixed

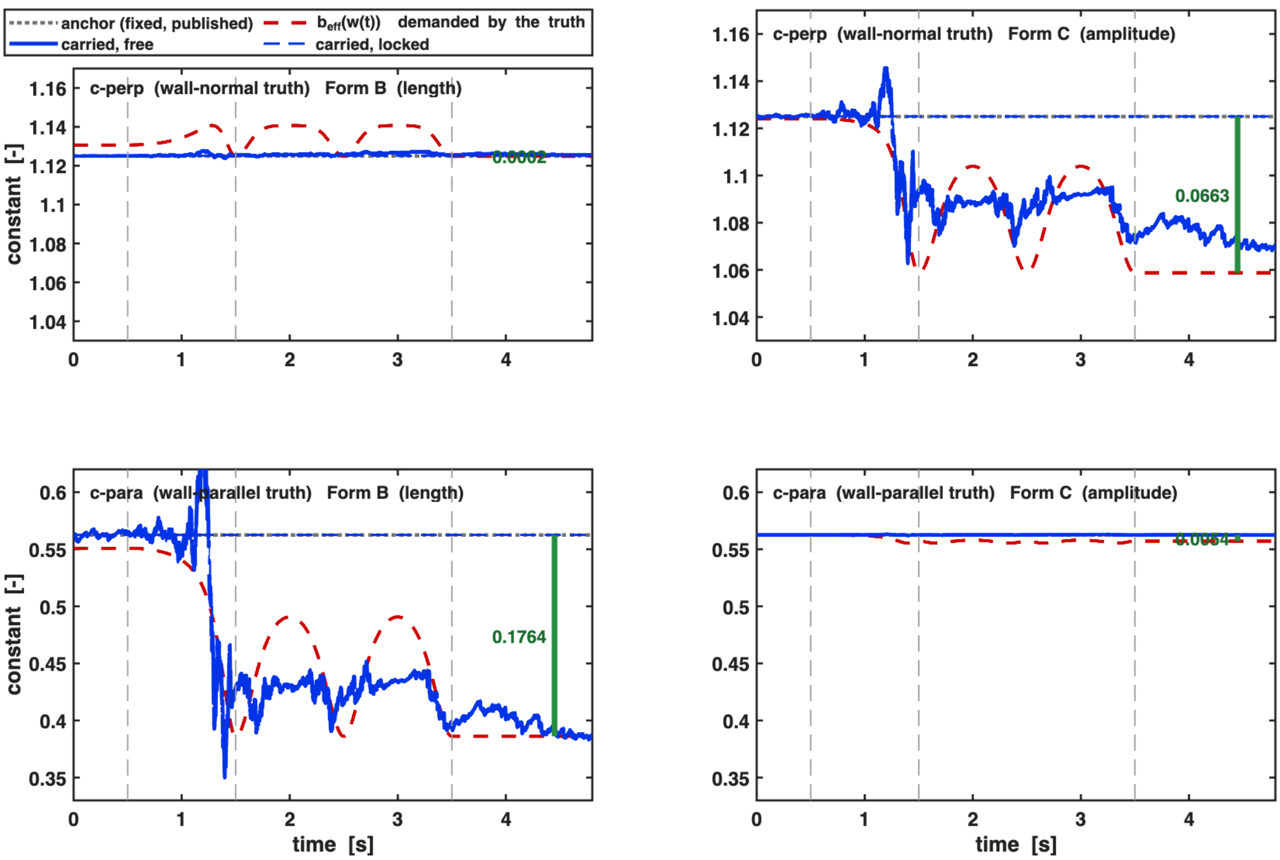
target $\quad b_{\text{eff}}(\bar{w}) = \text{page } 1$ $\qquad$ a curve: what the truth demands at each height

truth $= c_\perp$ and $c_\parallel$; $\qquad p = 1,\ \bar{w}_s = 1$ pinned; $\qquad$ 12 seeds

truth	writing	b_0	b_{eff} (hold)	$\lvert b_0 - b_{\text{eff}} \rvert$	value of freeing it
$c_\perp$	B	1.1250	1.1248	0.0002	$+0.084\ \ (t = +1.63)$
$c_\parallel$	C	0.5625	0.5571	0.0054	$-0.016\ \ (t = -0.68)$
$c_\perp$	C	1.1250	1.0587	0.0663	$-1.610\ \ (t = -2.92)^\dagger$
$c_\parallel$	B	0.5625	0.3861	0.1764	$-4.064\ \ (t = -6.48)^\dagger$

value $:= \lvert \overline{e_a} \rvert_{\text{free}} - \lvert \overline{e_a} \rvert_{\text{locked}}$ [%], $\quad$ final hold; $\qquad$ slope of the two large rows : $-24.3,\ -23.0$

locked pins the shape constant only (slot 5): its Kalman gain and covariance are zero and it never leaves b_0. The gain state $\bar{a}$ is estimated in both arms, so this is the marginal value of *also* freeing the shape constant. † these two rows share a sign with the $P(4,5) \to P_{44}$ inflation channel and await its control arm; the other two do not.

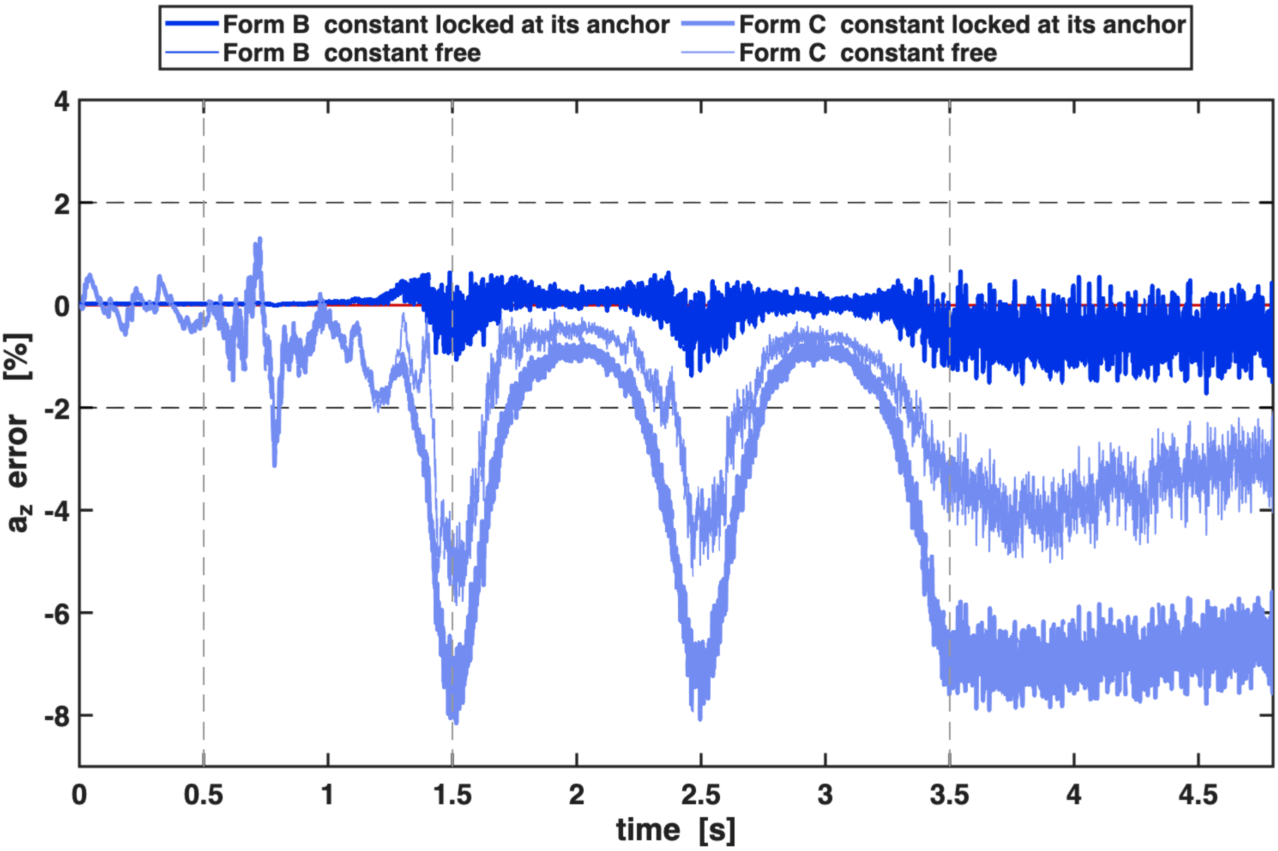


$$\text{truth} = c_\perp, \qquad b_0 = \tfrac{9}{8} \text{ for both writings}, \qquad \sqrt{P[0]} = 0.0157 \text{ (B)}, \quad 0.0708 \text{ (C)}$$

writing	constant	descent peak %	oscillation RMS %	final hold %
B	locked	1.032 ± 0.151	0.341 ± 0.075	-0.482 ± 0.368
B	free	1.102 ± 0.219	0.407 ± 0.144	-0.470 ± 0.579
C	locked	7.027 ± 1.953	2.762 ± 1.205	-5.579 ± 2.328
C	free	6.605 ± 2.652	2.389 ± 1.266	-3.083 ± 3.471

$$\text{locked} \;\; \text{B/C} = 0.15,\; 0.12,\; 0.09 \qquad \Longrightarrow \qquad \text{B ahead by } 6.8\text{–}11.6\times$$

The locked pair is the writing-versus-writing comparison with no estimator in it. Page 1: $b_{\text{eff,B}}$ swings 0.0189 here, $b_{\text{eff,C}}$ swings 0.0699.



truth $= c_\parallel$ substituted into the wall-normal channel, $\qquad b_0 = \frac{9}{16}$ for both writings

writing	constant	descent peak %	oscillation RMS %	final hold %
B	locked	10.914 ± 1.369	4.572 ± 1.177	-9.238 ± 1.789
B	free	9.728 ± 2.624	3.059 ± 1.560	-3.972 ± 4.246
C	locked	0.790 ± 0.152	0.375 ± 0.138	-0.506 ± 0.201
C	free	0.783 ± 0.190	0.373 ± 0.171	-0.490 ± 0.279

$$\text{locked}\ \ \mathrm{B/C} = 13.8,\ \ 12.2,\ \ 18.3 \qquad \Longrightarrow \qquad \text{the order of page 7 is reversed}$$

b_{eff} swings 0.1769 (B) against 0.0069 (C) here, the mirror of page 7. Form B pins a simple pole at $\bar{w}_s$; $c_\perp$ has exactly that pole (residue 1), $c_\parallel$ as implemented has none (finite 3.084 at contact). Two limits on reading this page: $c_\parallel$ is a substituted truth, not the x/y axes; and at *equal* degrees of freedom — Form B with $b = \frac{9}{16}$, p pinned and the origin free — the shape residual is 0.134 % against Form C's 0.404 %, so the ordering above holds under the rule that no constant may be fitted.